SWARMRIDE BHARAT · SELF-AWARE TRAINS FOR INDIAN RAILWAYS

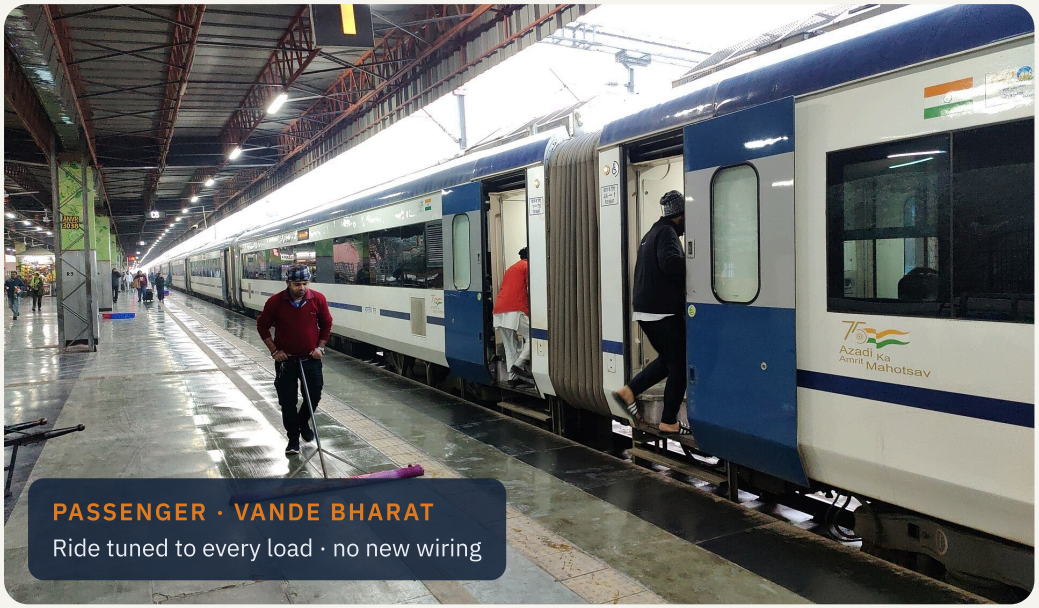
Every bogie powers itself, rides smarter and checks itself.

WHAT WE DO

We replace the oil damper on each bogie with a **smart magnet-and-coil damper**. The bump energy that becomes heat today now **powers safety sensors** and **tunes the ride**.

APPROACH

1. **Harvest** — Smart damper makes power
2. **Store** — Power bank on every bogie
3. **Sense + tune** — Axles, track, load, ride
4. **Sync** — Loco warns every coach



PASSENGER · VANDE BHARAT
Ride tuned to every load · no new wiring



GOODS · BOXN WAGONS
Every axle watched · wagons power themselves

~3 / day
hot-axle cases on Indian Railways
Railway Board, 2025

327
derailments from track faults, 2017–21
CAG audit

30–100%
shortfall in Track Recording Car checks
CAG audit

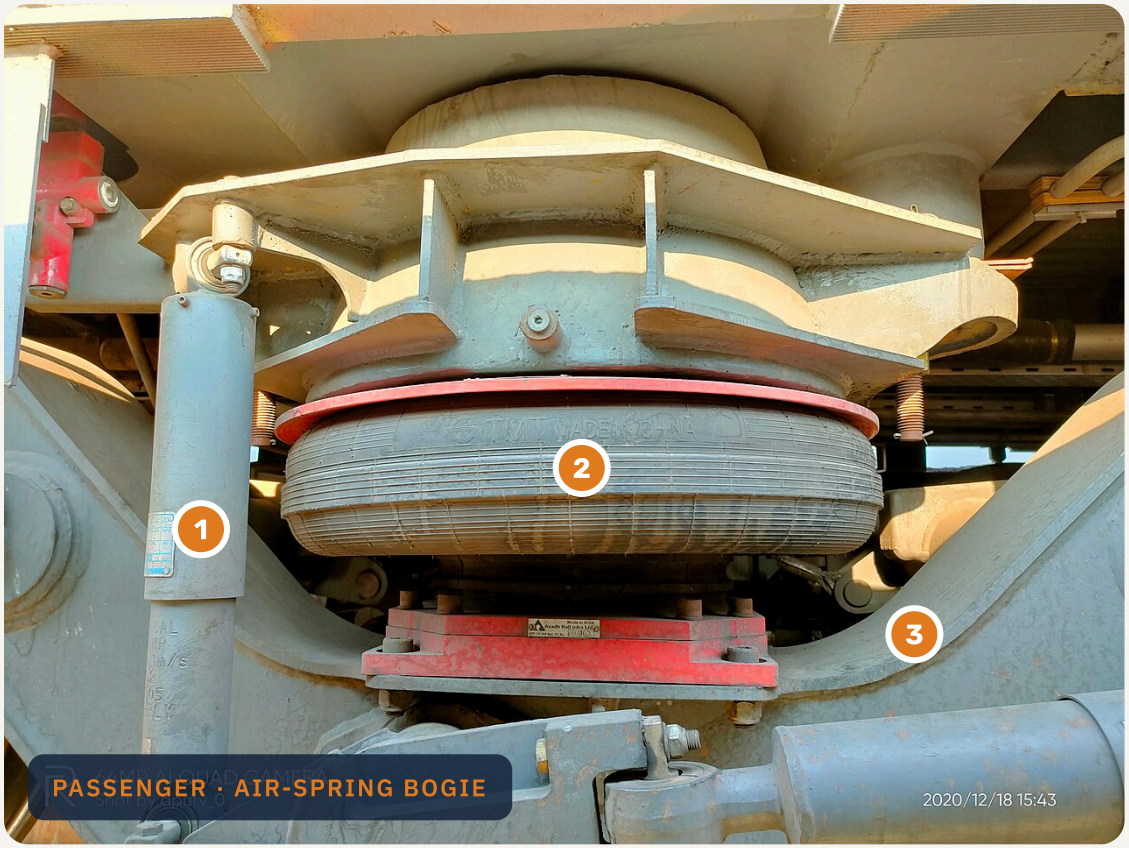
0 W
electrical power on a goods wagon today
no sensors possible

The energy is already in every bump. Today it becomes heat. We turn it into safety and comfort.

Photos (Wikimedia Commons): Vande Bharat Express — Ravi Dwivedi, CC BY-SA 4.0 · BOXN freight train at Rajahmundry — Adityamadhav83, CC BY-SA 3.0

WHERE IT FITS · REAL BOGIES, REAL PARTS

Same bogie. We swap one part and add a small power bank.



1 **Oil damper today → SwarmRide smart damper.** Same mounts. Moves with the suspension and makes power.

2 **Air spring stays.** Its pressure gives the live passenger load.

3 **Bogie frame → sealed power bank + controller.**



4 **Secondary springs + oil damper** → smart damper fits here. 5 **Axle box** → bearing sensor.

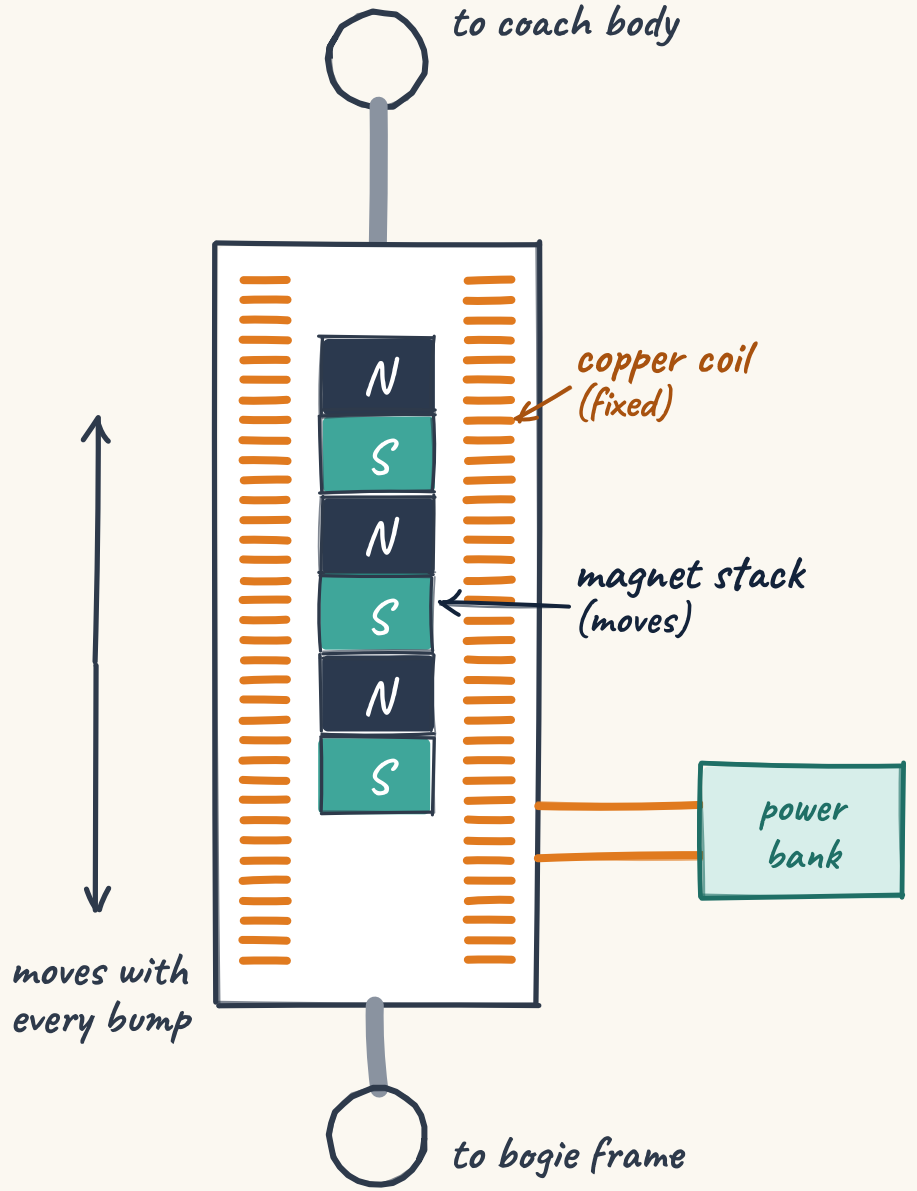


6 **Spring nest** → compact smart damper beside it. 7 **Axle box** → bearing temperature + vibration sensor.

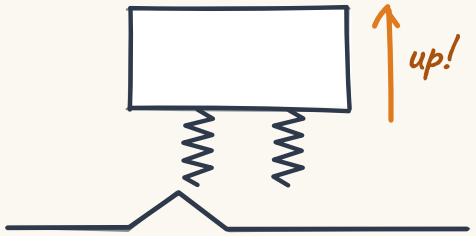
No new bogie • no new cables • fits today's mounts

Photos: air spring on LHB coach — Apurv013, CC0 (Wikimedia Commons) · ICF coach bogie and Casnub wagon bogie — supplied by the author

A magnet moving through a coil: power and ride control in one part.

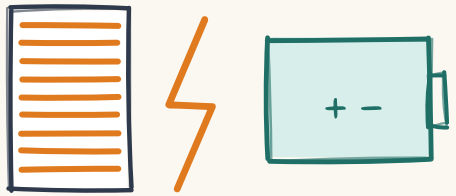


1 · A bump moves it



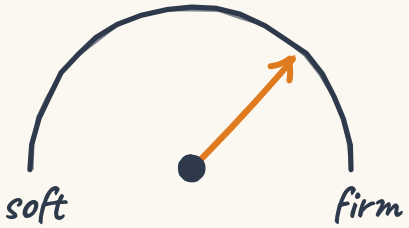
Track bump → coach moves → magnet slides through the coil

2 · It makes power



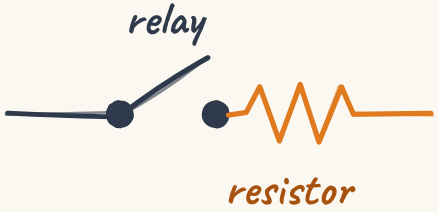
e = Ke · v (Faraday's law) → charges the bogie power bank

3 · It sets the ride



c = Ke² ÷ (R coil + R load) → firmness set in milliseconds

4 · It fails safe

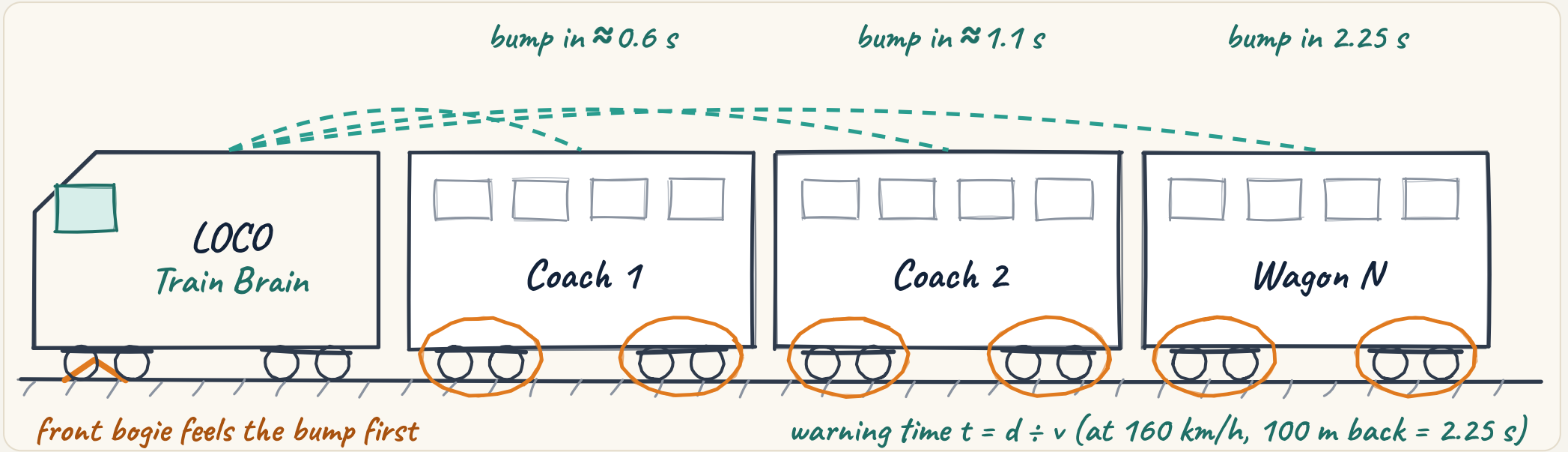


Power lost → relay → resistor → works like a normal damper

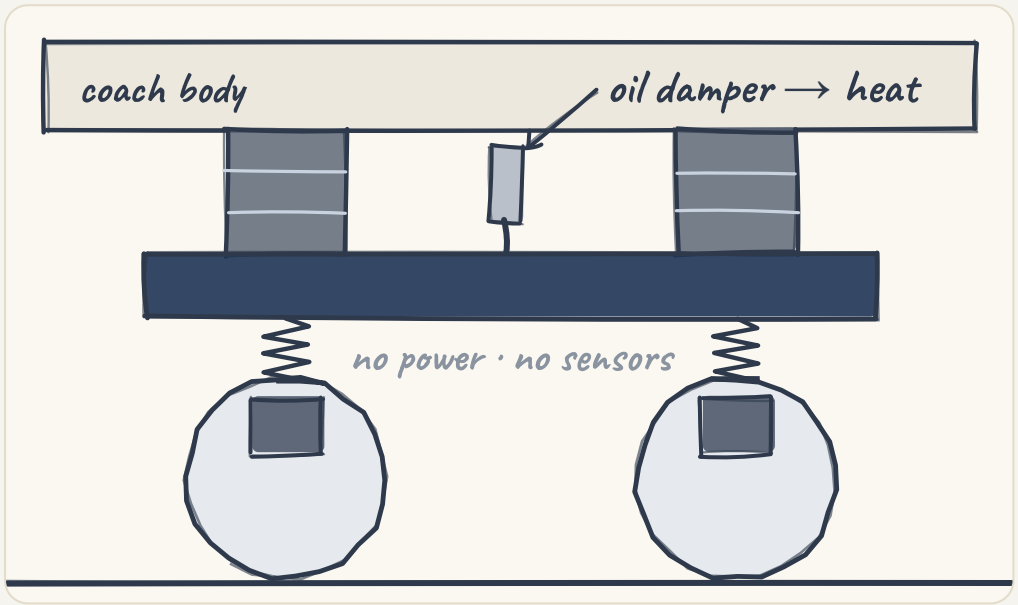
Fits existing damper mounts • no oil • no gears

Concept sketches, not to scale

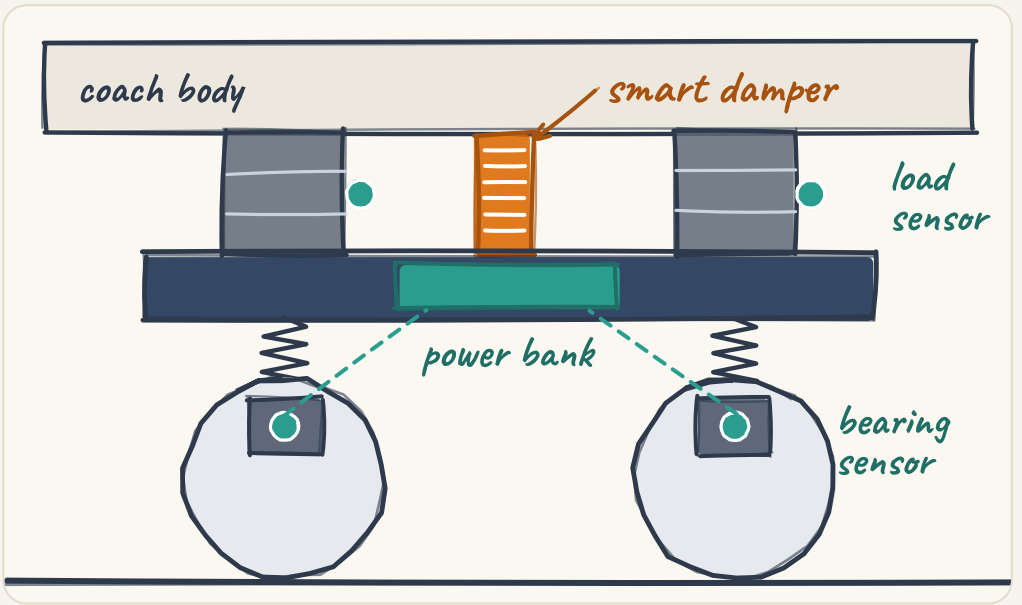
The loco warns every coach before the bump arrives.



Bogie today



SwarmRide bogie



Orange rings: SwarmRide on every coach and wagon bogie · concept sketches, not to scale

TWO PRODUCTS · ONE TECHNOLOGY

Comfort for passenger trains. Safety for goods trains.

SwarmRide Passenger

Vande Bharat · LHB · ICF



- Ride tuned to live load from air-spring pressure
- Ready before each bump: warning from the loco
- Track recorded on every trip
- No new wiring: each bogie powers itself

SwarmRide Freight

BOXN · DFC rakes



- Every axle: bearing temperature + vibration
- Train parting found at the exact wagon
- Track recorded on every goods run
- Wagons have no power; now they make their own

RIDE	HOT AXLES	PARTING	TRACK CHECKS	BOGIE POWER
one load	~25 checkpoints	last wagon	periodic	none
↓	↓	↓	↓	↓
every load	**every axle, always**	**exact wagon**	**every train, daily**	**self-powered**

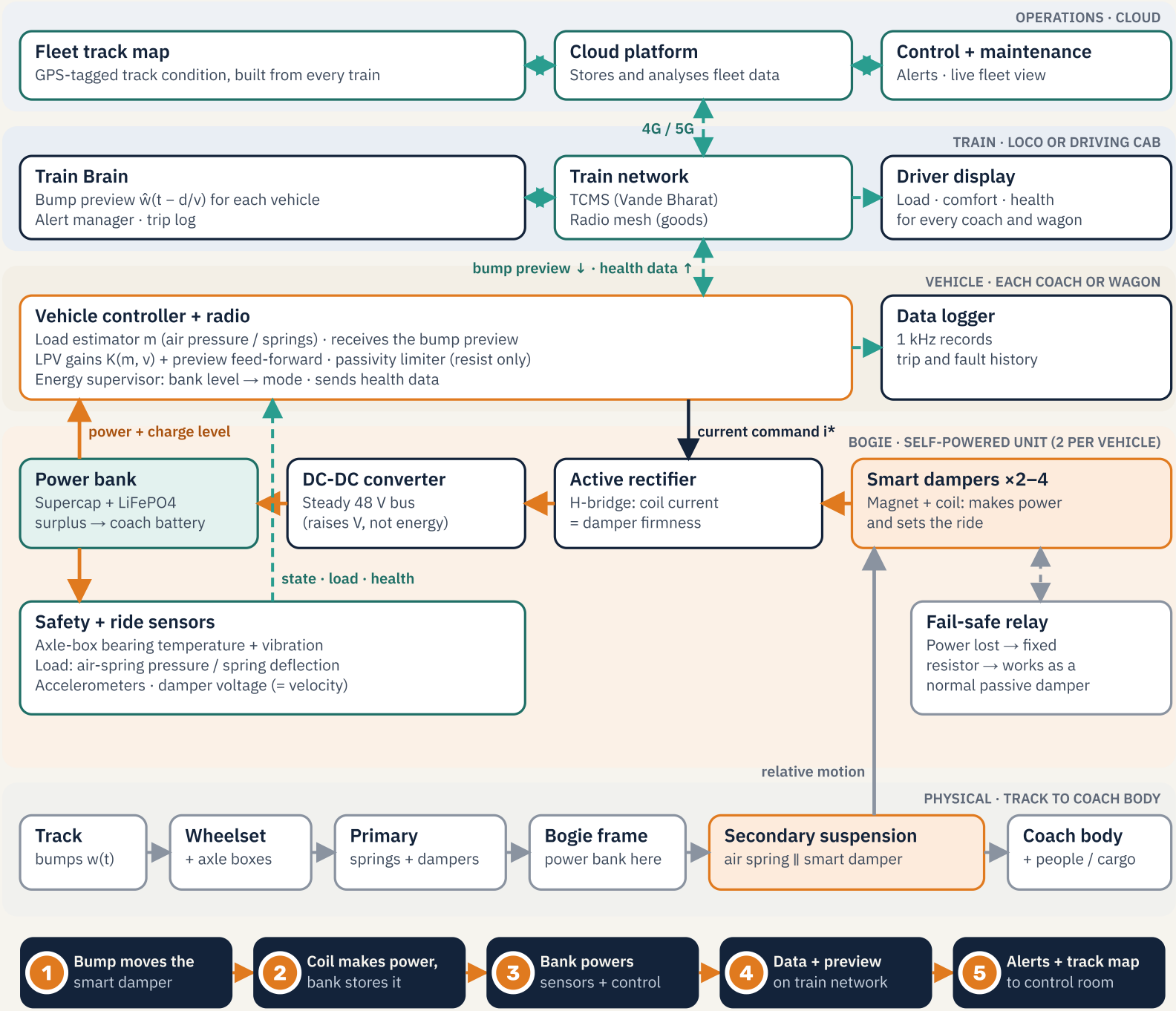
Prove it · simulation + bench → **RDSO approval · trials** → **Pilots · DFC rake + Vande Bharat** → **Fleet rollout · factories + data**

One damper. Two markets. Every bogie powers itself.

Photos: ICF coaches — supplied by the author · BOXN freight train — Adityamadhav83, CC BY-SA 3.0 (Wikimedia Commons) · expected results, to be proven

END-TO-END ARCHITECTURE

From a bump on the track to the control room.



LEGEND

- Harvested power
- Data link (arrows show direction)
- Control command
- Mechanical motion

ENERGY ORDER ON EVERY BOGIE

1. Safety sensors
2. Radio link
3. Ride control
4. Reserve for stops
5. Surplus to coach battery

CONTROL LAW

$F^* = -K(m,v)\cdot\hat{x} + Kp\cdot\hat{w}$

$c = \text{clip}(F^* \div v_{rel}, 0, c_{max})$

$i^* = c\cdot v_{rel} \div Kf$

Resists only: every bit of damping is harvested energy.

WORKS ALONE

If the train network drops, each vehicle keeps load-adaptive control without the preview.

SAFETY

Fail-safe relay makes it a normal damper. Safety never depends on software. To be tested to EN 50155 · EN 61373 · EN 50121.

PROVEN IN SIMULATION · MATLAB / SIMULINK · FOUR STAGES

Same coach, same rail, only the damper changes. SwarmRide wins.

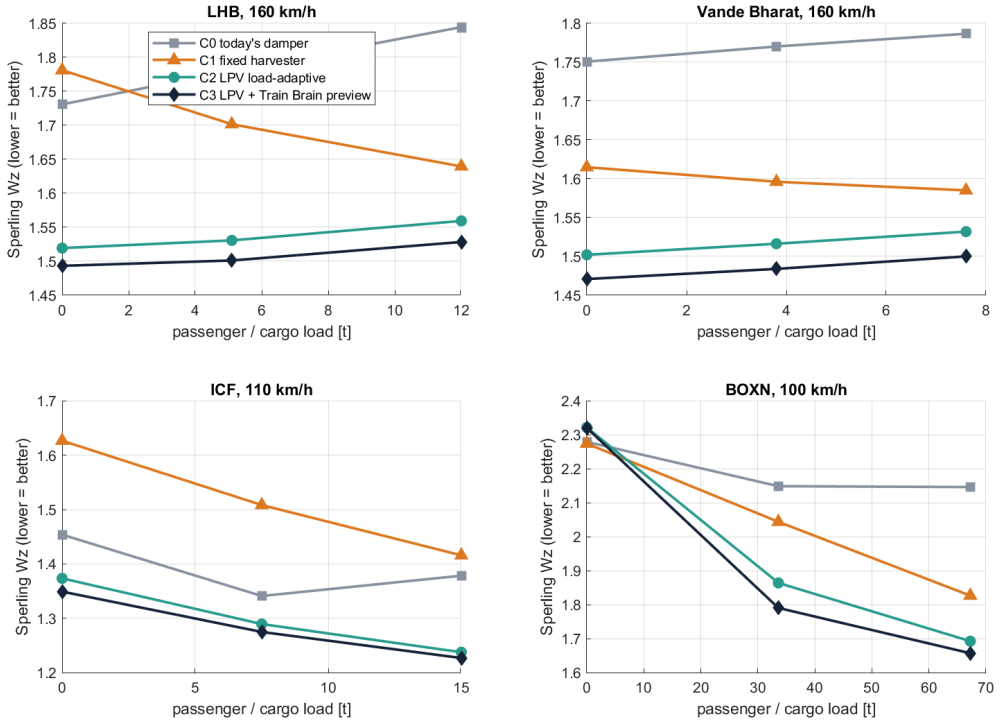
−17%
ride index, crowded LHB coach at 160 km/h (Vande Bharat −16%, loaded wagon −23%)

−41%
body vibration on the crowded LHB coach; suspension travel down by a third

33–38 W
into each coach bogie's bank on rough track, against 8 W of electronics

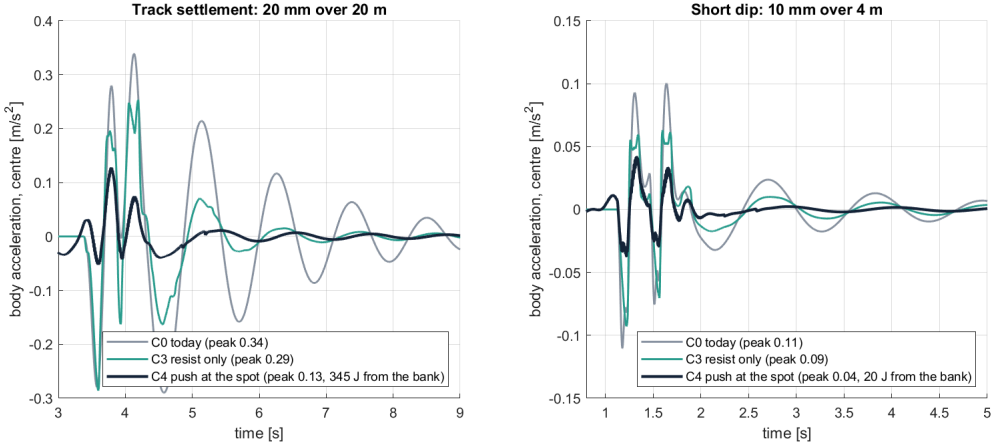
−80%
worst-seat jolt at a track settlement when the damper pushes, for 0.1 Wh

Ride comfort at every load (centre of the coach, poor track)



Better at every load. Today's ride worsens as passengers board; SwarmRide stays nearly flat.

Active push at known bad spots: the Train Brain warning finally pays off (LHB crowded, 160 km/h)



The Train Brain warning pays off. At a known bad spot the damper briefly pushes with banked energy.

All day, all night

More made than used on LHB, Vande Bharat and freight, with a 1 W sleep when parked

Hot axle in 10 s

On-bogie sensor; trackside checks miss 82% of faults until after 180 °C

Wheel defect ×5–×200

Seen in the damper's own coil signal, located to the wheelset

Verified: hand calculations, an independent model, energy conservation, step-size convergence, fail-safe and controller stability in an automatic test suite. **Honest limits:** vehicle parameters are estimates pending RDSO data; vertical model only; the bogie electronics must sleep when parked to stay self-powered overnight.